

Spark Daemon Pattern: review and decision package (r4.4)

- **Status:** PROPOSED FOR INDEPENDENT REVIEW. NOT ADJUDICATED. It authorizes no LTC version change, no daemon class unlock and no activation. Where it recommends amending a settled ruling, the owner decides.
- **Date:** 2026-09-30. Prepared by Claude for the owner, to hand to other reviewers (people or models).
- **Repository state:** spark-daemon-pattern/ at revision r4.4, on branch `claude/clever-bardeen-qqkxez`:
 - 205 of 205 self-tests pass;
 - the four reference daemons pass 18 of 18 conformance-battery checks on the build workspace (Landlock ABI 7);
 - the daemon contract is 3.0.0.
- **This document stands alone.** It combines the review handoff (`HANDOFF-REVIEW-PACKAGE.md`) and the decision brief (`DECISION-BRIEF.md`). Every claim names the file in the repository that holds the detail, for reviewers who can read it. A reviewer without the repository can still mark every item.

What is inside:

- **Reviewer focus (read first).** For every decision: what is built and where, what remains, and the new seams the remaining work would create.
- **Part 1: the review handoff.** What the pattern is and what is built; the owner's settled rulings; the three review packages and their decision registers:
 - Package A: the Local Transport Contract;
 - Package B: Break Glass and operator-gated recovery;
 - Package C: other open decisions.
- **Part 2: the decision brief.** The settled rulings revisited (none reversed; one amendment or clarification each), and every open decision with its options, pros, cons and a recommendation, in a suggested order.
- **Part 3: one markup table** for everything above, and the document map.

Section numbers restart in each part; "Part 2 §2.4" means section 2.4 of Part 2.

Reviewer focus: what is built, what remains, and the new seams

Read this section first. It says, for every decision in this package:

- what is implemented today and where in the code;
- what remains to be built;
- which **seams** the remaining work would create.

A seam is a new boundary where two parts meet and each assumes something about the other.

Why the seams deserve the most attention. Every defect found since r3.5 sat at a seam, where two limits or two components were each correct on their own and wrong together:

Defect	The seam it sat on
HF-28	The runtime's exit codes and the unit's restart policy: <code>SENSE_BLIND</code> was restarted forever
HF-30	A daemon's worst-case sense time and its watchdog period
HF-31	The unit's stop timeout and the length of one cycle
HF-32	Process restarts and the blind clock

Defect	The seam it sat on
HF-33	The skeleton's ctx API and daemon authors: a truncated listing was returned as a value and diffed as complete

Reviewers should test each seam below with the question: "What does each side assume about the other, and what happens when that assumption is false?"

Implementation status

Status labels:

- **Built:** in code, with tests.
- **Partial:** some of it is in code.
- **Designed:** recorded in documents only.
- **Proposed:** new in this package.

Code paths are under spark-daemon-pattern/.

Decision	Status	What exists today (where)	What remains
PD-63 Blind period (settled)	Built (r3.5)	ctx.unsettled (spark_daemon/context.py); blind_limit_seconds (spark_daemon/manifest.py); the blind clock _Blind and SENSE_BLIND exit 78 (spark_daemon/runtime.py); RestartPreventExitStatus=2 65 73 78 (spark_daemon/unitgen.py)	Proposed amendment: split exit 78 between EXIT_POLICY and EXIT_SENSE_BLIND (spark_daemon/__init__.py, the unit, the contract's exit table). Designed: where T is set, deferred to PD-40
PD-70 Blindness across restarts (settled)	Built (r4.0)	_AcceptEvidence reads the verified ledger at start-up; DAEMON_HEARTBEAT every T/2; one reacquisition cycle (spark_daemon/runtime.py); four tests (tests/test_blind.py)	Proposed amendment: measure on CLOCK_BOOTTIME within a boot (boot_id and boot time in the heartbeat, start and stop records); assume the worst across boots; add a test for a backward clock step
PD-01.9 Blind Forester (settled)	Partial	The contract names the slot: fail closed for observers (spark_daemon/contract.py, cycle.blind_forester)	Everything for active classes: the survival loop in the supervisor (BF-1), marked outputs (BF-2), validity windows (BF-3), pre-authorized actions (BF-4), recovery exit (BF-6). Blocked on PD-72. Proposed: the scope clarification
PD-76 Direct memory reading (settled)	Designed; feasibility measured	DB-14 still decides on a per-process peak (spark_daemon/battery.py, db14). The feasibility probe is in evidence/r4.4/cgroup_direct_peak.py	DB-19: spawn the daemon inside a fresh cgroup before it runs, read v2 memory.peak or v1 memory.max_usage_in_bytes, record the version, retire DB-14. The CI rule for INCOMPLETE
HF-33 Capped listing	Built (r4.3), for dir-watch only	A listing over the cap is a failed cycle (examples/dir-watch/daemon.py); three tests (tests/test_blind.py::DirWatchCapacityTests)	The framework-level rule is PD-83
PD-82 Bind, do not embed	Proposed	Nothing	The manifest field for declared connections; register fields; a battery hook for the LTC suite

Decision	Status	What exists today (where)	What remains
PD-83 Truncated reads raise	Proposed	ctx.read_text already raises (spark_daemon/context.py). ctx.run, ctx.git and ctx.list_dir return truncation as a value	Change the three methods; add an opt-in partial result type; contract 4.0.0
PD-84 Register vocabulary	Proposed	The battery report binds the manifest, contract and code digests (spark_daemon/battery.py)	The PD-40 register itself, with LTC binding fields
PD-85 Declared pacing	Proposed	Uniform random jitter, $\pm \min(5\%, 30\text{ s})$, zero in test mode (spark_daemon/runtime.py, _jitter_bound)	A pacing manifest field; deterministic spread by default
PD-86 Retry bounds over time	Proposed	The unit's start limit, 5 in 300 s (spark_daemon/unitgen.py)	A lifetime bound per retrying layer; a battery cross-check
PD-01.10 to PD-01.14 Break Glass and recovery	Designed (BREAK-GLASS.md)	Nothing	All of it. Blocked on PD-72, PD-01.7, PD-40 and a heartbeat acknowledgement path
BF-1 to BF-6	Designed	—	See PD-01.9
PD-72 Supervisor/worker split	Designed (ADJUDICATION-AP.md, AP-04)	Nothing	A supervisor process with its own Landlock domain; a per-cycle worker deadline
PD-40 Activation register	Designed	Candidate envelopes and battery reports carry the digests it would bind	The register, revocation (U-5), evidence validity periods (U-7)
PD-01.3 Sentinel	Designed	Heartbeats already expose mode: blind	SENSE_DEGRADED at T/2, spark-escalation/1, a read-only escalations command
PD-69 Limits checked against each other	Partial	Individual cross-checks are tests: the git-watch sense budget against its watchdog, the stop timeout against one cycle, the restart policy (tests/test_blind.py)	The per-daemon budget table in DB-15 (Phase 1)
PD-71 (remainder)	Designed	One small fixture per daemon	Named worst-case fixtures; a scaling series to derive N_max
C-1, PD-75 to PD-81	Designed	The battery exits PASS 0, INCOMPLETE 3, FAIL 1 (spark_daemon/__init__.py, EXIT_FLAGGED = 3)	The code change with contract 4.0.0; importing and reconciling the v0.1 line's registry (PD-78)
Independent ledger verifier (B-6, U-7)	Designed	DB-04 and DB-16 use the skeleton's own verifier	A verifier that does not import spark_daemon
Phase 1	Designed	—	An OnFailure= notifier; a diverse staleness checker; a budget report

New potential seams

Each seam is introduced by the decision named. Each entry states the assumption that could fail and what to check.

#	Seam (introduced by)	The assumption that could fail	What a reviewer should check
N-1	Supervisor and worker (PD-72, BF-1)	The supervisor keeps a survival loop running while the worker is blind or dead. But it shares the host, and under LTC-14 it must not fabricate the worker's results	Are survival outputs the supervisor's <i>own</i> , marked records? Does exit 78 still mean "never restart" for the worker, while the supervisor never exits blind? Does each side have its own Landlock domain?

#	Seam (introduced by)	The assumption that could fail	What a reviewer should check
N-2	Daemon and activation register (PD-40, PD-49, PD-84, PD-63's option B)	The register binds exactly what the daemon runs	Which digest is compared (file bytes or canonical JSON)? Is a trust result checked for age and revocation? If T moves to the register, is its value recorded in DAEMON_START?
N-3	Declared connection and LTC provider (PD-82, LI-2, LI-11)	The manifest's bounds and the LTC profile's bounds are the same numbers	Is there one source for each bound (U-8)? Does the battery run the LTC suite against the <i>bound</i> provider?
N-4	Battery and cgroup (PD-76, DB-19)	The daemon is inside the cgroup from its first instruction, and readings from cgroup v1 and v2 are comparable	Is the daemon placed in the cgroup before exec, as the probe does? v1 counts page cache differently, so is the report's cgroup version compared only with a limit for the same kind of host? The battery gains a privileged step: what if it fails halfway?
N-5	Ledger and the external staleness checker (Phase 1, A-2, U-7)	"The ledger file's age" measures liveness	File age compares a modification time with the wall clock, so it has the same clock-step weakness as PD-70. Prefer a checker that notes the file's size or hash and checks for change on its <i>own</i> monotonic clock
N-6	Heartbeat and orchestrator acknowledgement (Break Glass trigger, BG-1)	An acknowledgement is only ever read	This is the first <i>inbound</i> path into a daemon that is pull-only today. It must never carry commands, and its absence must be one of two witness signals, not a trigger on its own (BG-12)
N-7	Break-glass relay and external system (BG-5, BG-10)	Every command to the target passes the one gate that enforces "highest epoch wins"	Is <code>single_path: true</code> verifiable, or only declared? What does the external system do with an out-of-epoch command that reaches it by another path?
N-8	Witness and supervisor (BG-12, BG-13)	The two signals are independent, and the epoch copies agree	Do the two signals share a clock, a network link or a host? On a single DGX, is a separate process tree enough?
N-9	Flight recorder and daemon ledger (BG-15 to BG-17)	Two separate chains can be reconciled and verified independently	Is the seal's head hash anchored somewhere the supervisor cannot rewrite? Is the independent verifier used for both chains?
N-10	The ctx API and daemon authors (PD-83)	An opt-in partial result keeps its flag all the way into the records built from it	Can the flag be lost by copying fields into a new dictionary? Should the record builder refuse a partial value that has lost its marker?
N-11	Pacing and watchdog (PD-85)	A phase offset never delays a watchdog ping	The runtime pings the watchdog during the sleep, independently of the offset. Confirm this holds for FIXED_PHASE and for offsets close to the interval
N-12	One exit-code table and its four consumers (PD-63's split, PD-35, C-1)	The runtime, the unit's RestartPreventExitStatus, an OnFailure= notifier and the Observer's table agree	Is there a single table from which all four are generated, and a test that fails if one of them drifts?
N-13	Contract 4.0.0 and existing manifests (the batched change)	Every author migrates once	Is there a schema-3 migration hint, like the one from version 1 to 2? Do the battery check IDs change meaning (PD-78: retire and replace, never edit in place)?

#	Seam (introduced by)	The assumption that could fail	What a reviewer should check
N-14	Survival mode and emergency authority (PD-01.9 and PD-01.12)	A restart resumes survival mode (PD-70) and ends break-glass authority (BG-9)	Can any record written in survival mode be read at start-up as authority? The two must use different record types

Part 1: Review handoff

0. How to review

1. **Mark each item** in the decision registers (Part 1 sections 3.5, 4.4 and 5, and Part 2 sections 1 to 3): **APPROVE**, **MODIFY**, **REJECT** or **DEFER**, with notes. Part 3 has one blank markup table for both parts.
2. **Owner rulings are settled** (section 2). Do not reopen them. Their *implementation conditions* are open, and are marked as such.
3. **The three packages are separate.** Review them separately, and do not move an identifier from one to another.
 - **Package A:** the Local Transport Contract (LTC) and the daemon pattern.
 - **Package B:** Break Glass and operator-gated recovery.
 - **Package C:** other open decisions.

The LTC hardening markup requires that Break Glass "remain a separate review package and not silently consume" LTC placeholder numbers. This handoff keeps that rule.

4. **Evidence standard.** A finding counts when it is reproduced, measured or traced to code. Anything else is recorded as an opinion. Say which yours is.
5. **Identifier namespaces**, so that nothing collides:

Prefix	Owner	Meaning
PD-nn, PD-01.n	This daemon pattern	Pending decisions. PD-75 to PD-81 are aliases of the parallel v0.1 line's PD-23 to PD-29
HF-nn	This daemon pattern	Defects found, reproduced and fixed, each with a test
S-n, U-n, BM-n, BF-n	This daemon pattern	Safety invariants, universal-contract rules, blind modes, Blind Forester conditions
BG-n, OR-n	This daemon pattern (Package B)	Break Glass and recovery requirements. Not LTC numbers
LI-n	This daemon pattern (Package A)	LTC integration requirements. Not LTC numbers
LTC-nn, LTC-V2-nn, CT-nn	The LTC	The LTC's own requirements and conformance tests
LTC-H01 to H03	The LTC hardening markup	Placeholders until the LTC adjudication assigns numbers

1. What the daemon pattern is, and what is built

A **daemon** is a long-running Spark process. The pattern has three parts:

- **A closed-schema manifest** declares everything about the daemon's safety: what it reads, which commands it may run, resource limits, watchdog, blind limit and ledger rules.
- **A fixed skeleton** supplies every safety mechanism. The author writes only sense (read), decide (pure) and digest (pure). The skeleton provides:
 - a purity check;
 - a Python audit hook;

- Landlock file-system and network confinement;
- a generated systemd unit with a sandbox, watchdog and restart policy;
- a hash-chained, append-only JSONL ledger (JCS canonical form, one fsync per record);
- a bounded Markdown digest.
- **A conformance battery** runs the real daemon through 18 checks (DB-01 to DB-18) before anyone may activate it.

Only class 1a (observe and record) exists. Activation is a human Class C decision; a battery PASS is evidence for it, not permission. Nothing here installs or starts a service.

Built since the r3 baseline (all tested):

- **32 defects fixed, each with a regression test** (HF-01 to HF-33; HF-23 is a missing-evidence note, not a defect), including two confinement escapes and a Git code-execution path. Three (HF-28, HF-30, HF-31) were found by reading and arithmetic; the rest were reproduced by running.
- **The blind period.** A daemon that accepts no observation for `blind_limit_seconds` (T) stops with `SENSE_BLIND` (exit 78) for a human, and 78 is never restarted.
- **Blindness survives restarts** (PD-70). The clock starts at the ledger's latest evidence of an accepted cycle. A `DAEMON_HEARTBEAT` is written every T/2, and one reacquisition cycle is allowed per restart.
- **A published, versioned daemon contract** (3.0.0), and candidate envelopes that carry no authority.

Designed, not built:

- the supervisor/worker split (PD-72);
- the activation register (PD-40);
- the connector gate (PD-01.7);
- every class above 1a;
- everything in Packages A and B except HF-33.

Known limits of the evidence:

- **Never run under systemd as PID 1.** The unit's directives are checked by generation and lint, not by a real service manager.
- **Landlock ABI 7 comes from the build workspace.** The DGX's ABI is inferred (ADJUDICATION-AP.md).
- **No test ran on DGX hardware.**

2. Owner rulings already made (settled)

Do not reopen these here. The owner asked separately whether any is worth revisiting. Part 2 §2 answers that: none should be reversed, and each has one proposed amendment or clarification:

- PD-63: separate exit codes for a policy violation and `SENSE_BLIND`;
- PD-70: measure inherited blindness on `CLOCK_BOOTTIME` within a boot;
- PD-01.9: state that it does not unlock `critical`, and make BF-5 mandatory;
- PD-76: accept a direct reading from either cgroup version (measured: 208 MiB for the cgroup against DB-14's 128 MiB).

Review those amendments there.

Ruling	What it settled	Still open for review
PD-63 (2026-09-29)	The blind period is in force, as built	—
PD-70 (2026-09-30)	Heartbeat every T/2; blindness counted across restarts. Built with one reacquisition	—

Ruling	What it settled	Still open for review
	cycle	
PD-01.9, "Blind Forester" (2026-09-30), adopted as core pattern	An <i>active</i> daemon whose stopping would drop a critical downstream payload shifts into a pre-validated degraded survival loop instead of failing closed. For observers, the survival action remains "fail closed"	BF-1 to BF-6 , the implementation conditions: the loop runs in the supervisor; survival is signalled; last-known-safe outputs have a validity window; survival actions are pre-authorized; the far side keeps its own procedure; leaving survival mode requires revalidation
PD-76 (= the v0.1 line's PD-24(a)), adopted	Memory for the budget check needs a direct cgroup memory .peak reading. RSS-only runs are INCOMPLETE	How CI treats INCOMPLETE where the reading cannot be taken (Package C)

3. Package A: the Local Transport Contract and the daemon pattern

Source documents:

- *Spark Core Local Transport Contract v0.1 + v0.2 Upgrade Requirements* (LTC-01 to LTC-15, CT-01 to CT-26, LTC-V2-01 to LTC-V2-20);
- *LTC Hardening Review: Reviewer Markup Handoff* (LTC-H01 to LTC-H03).

Detail: LTC - INTEGRATION - REVIEW.md.

3.1 The question

Should a standardized transport contract sit *within* the daemon pattern?

Proposed answer: bind, do not embed. The LTC defines how a component submits bounded local work to an authorized provider and receives bounded completions. Today's daemons only observe and append to their own ledger, so they have no LTC channel. The ledger is expressly *not* one: LTC-01 makes transport state ephemeral and non-canonical, while the ledger is the durable evidence layer.

The LTC becomes relevant the moment a daemon gains a *request channel*. That is how every class above 1a is designed to grow:

- an Analyst asking an inference engine;
- an Operator submitting a ticket to an executor;
- the connector gate as a provider;
- the break-glass relay as an alternate provider.

Option	For	Against
A. Embed the LTC in the pattern	One package, one version	Couples two version lines. Pulls daemon constants into the LTC. Observers carry unused transport code. The LTC is still a v0.1 candidate
B. Bind at declared edges (recommended)	One transport meaning across daemons, scripts and services. Gate conformance <i>is</i> the LTC suite. Fencing, backpressure and UNKNOWN_AFTER_RESTART defined once	Needs a small seam (manifest field, register fields, battery hook), designed before its first use
C. Keep them separate	Nothing to do now	Every class re-derives transport rules; the ad hoc borrowing already visible in PD-01 §6 continues

3.2 Evidence: the LTC-H01 rule found a real defect

LTC-H01's rule: "Exceeding any declared bound **MUST** produce a typed failure and **MUST NOT** yield a successful truncated payload."

HF-33 (Medium, reproduced, fixed in r4.3). The reference daemon `dir-watch` capped its folder listing at 512 entries. Past the cap, the skeleton returns the first entries *in directory order* and flags the result as truncated. The daemon diffed that partial listing as if it were the whole folder. Directory order shifts when a file is added, so each addition recorded some *other, unchanged* file as removed:

- 17 false removals in 20 additions on `ext4`;
- 20 in 20 on `tmpfs`;
- no truncation flag on the events.

The fix makes a folder over the cap a failed cycle, so it counts towards the blind limit and reaches a human. The tests fail on the previous revision (`evidence/r4.3/`, `tests/test_blind.py::DirWatchCapacityTests`).

The same shape caused HF-29 earlier: Git output over the cap was accepted as "unavailable". Two defects from one API design (truncation returned as a value that each author must remember to check) motivate PD-83.

3.3 Integration requirements (LI-1 to LI-14, all PENDING)

- **LI-1.** The ledger is not an LTC channel. Transport outcomes that matter are recorded in it as typed events; completions are observations, never authority.
- **LI-2.** A declared outward connection names an LTC channel profile: contract and version, schema, ordering scope, finite bounds, overflow policy, and a payload-handling profile.
- **LI-3.** The LTC binding is resolved at activation. The activation-register row *is* the LTC §12 binding output, and the daemon never consults a registry per message.
- **LI-4.** Compatibility (battery plus LTC conformance) and authorization (human activation) stay distinct.
- **LI-5.** The supervisor keeps to LTC-14. It never fabricates completions. **Consequence:** the Blind Forester's survival outputs are the supervisor's own marked submissions, never the worker's completions.
- **LI-6.** Channel incarnation (LTC-10) and break-glass authority epochs are separate fences. Both rotate at boot.
- **LI-7.** An alternate-provider switch (the break-glass relay) is a pre-bound, explicitly authorized rebind (LTC-V2-19). This belongs to Package B.
- **LI-8.** No skeleton read returns a truncated payload as success (PD-83).
- **LI-9.** Retry bounds hold over time, not per window (PD-86).
- **LI-10.** Pacing is a declared execution-profile field. The default for background observers is deterministic spread (PD-85).
- **LI-11.** The battery runs the LTC provider-neutral suite for each declared channel, and the report binds the LTC text's hash.
- **LI-12.** No constant crosses the boundary in either direction.
- **LI-13.** Telemetry (the heartbeat) never gains transport authority.
- **LI-14.** BG-, OR- and LI- identifiers are never used as LTC numbers.

3.4 This repository's markup on LTC-H01 to H03

Item	The markup's disposition	This review	Added
LTC-H01 Payload-handling profiles	APPROVE WITH MODIFICATION	Agree	(1) Profile fields are mandatory; NOT_APPLICABLE only explicitly, with a reason. (2) STREAM_BOUNDED needs a <i>total-operation</i> deadline, not only an idle one: start-up ledger verification here is bounded per record and never idle, yet grows without limit. (3) Test both sides of the boundary (exactly at the ceiling passes, one byte over fails). Test that a capped partial result offered as complete is rejected at consumption (HF-33's shape). (4) The typed failure appears in the operation's own outcome, not only in telemetry

Item	The markup's disposition	This review	Added
LTC-H02 Authenticated binding	MODIFY / SPLIT OWNERSHIP	Agree	(1) Trust evidence names the digest it covers, so evidence cannot be replayed onto another artifact. (2) A trust result has an age and a revocation state; offline verification reproduces both. (3) Name every digest by what it covers
LTC-H03 Retry and pacing	REJECT JITTER MANDATE; APPROVE BOUNDED RETRY	Agree, with a stronger rule	Per-layer, independently budgeted retry bounds are <i>not enough</i> : they must hold over an unbounded horizon. Evidence: HF-28 here had a finite, independent restart budget (5 per 300 s), yet a daemon failing every $T > 75$ s was restarted forever, because the retries were spaced wider than the window. Proposed extra test: retries spaced just outside the window must still terminate

3.5 Package A decision register

ID	Decision	Recommendation
PD-82	Bind, do not embed (option B, LI-1 to LI-7, LI-11 to LI-14)	APPROVE
PD-83	ctx.run, ctx.git and ctx.list_dir raise TooLarge on truncation by default; partial results only by explicit opt-in that carries the flag. A contract major (4.0.0), batched with other contract changes	APPROVE
PD-84	The activation register adopts the LTC binding vocabulary, with H02's three modifications	APPROVE
PD-85	Pacing is a declared field (NONE, FIXED_PHASE, DETERMINISTIC_SPREAD, BOUNDED_JITTER). The default is deterministic spread from the daemon's name and the host's identity. This amends PD-21, today's uniform random $\pm \min(5\%, 30\text{ s})$	APPROVE
PD-86	Retry bounds hold over time. Submit the H03 finding to the LTC review	APPROVE

Questions for reviewers:

- **A1.** Is "bind, do not embed" right, given that no daemon has a request channel yet? Or should the seam wait until the first one?
- **A2.** Is making truncation a typed failure worth a contract major (PD-83), or should the rule be enforced by the battery instead?
- **A3.** Deterministic spread gives a fixed phase per daemon, while random jitter random-walks. Is there a case where the fixed phase is worse for background observers?
- **A4.** Is the H03 strengthening ("bounded over time") correct and testable in provider-neutral terms?

4. Package B: Break Glass for external connections, and operator-gated recovery

Detail: BREAK-GLASS.md. Sub-decisions PD-01.10 to PD-01.14 in PD-01-DAEMON-CLASSES.md. This package is outside the LTC hardening review by that review's own scope guard.

4.1 The proposal and its constraints (owner)

The three mechanisms:

- Dormant Egress Relaxation;
- a Sovereign Survival Socket;
- Pre-Signed Capability Envelopes.

Then the owner added:

- emergency epochs with fencing;
- an independent two-signal witness;
- an Emergency Flight Recorder with mandatory reconciliation;
- an operator-gated recovery boundary: `SAFE_T0_ISOLATE`, a restart that opens a new epoch, and `REJOIN_PROBATION`.

Constraints:

- An AI model never decides that an emergency exists. "Do not let model judgment automatically confer state-mutating authority" (Governing Principle 10).
- The mechanisms apply only to connecting to *external* systems, never to internal spark-core systems.

4.2 Adjudication proposed

Item	Proposed verdict	Key reason
Reflex, not judgment	ADOPT (BG-1)	Only a fixed rule over mechanical signals starts an emergency; no model output is an input
External only	ADOPT, enforced (BG-2 to BG-4)	The validator refuses internal targets, and the relay re-checks the resolved address at connect time. Break glass never touches Spark's own confinement, control plane or authentication
Dormant egress	ADOPT MODIFIED (BG-5)	Landlock can only be narrowed, and its network rules cannot name a host. So the egress belongs to a separate, pre-granted, dormant relay that stops at fence
Survival socket (raw, unauthenticated, /tmp)	REJECT as first written; ADOPT the owner's revision (BG-6)	/tmp can be squatted, an unauthenticated socket is open to any local process, and it contradicts "authentication never degrades". The revision: exists only during an emergency epoch, lives in /run/spark-emergency/, identifies the peer by <code>S0_PEERCREATED</code> , has a closed command set, requires a hardware key for a technician at the machine, and is removed at fence
Pre-signed envelopes	ADOPT with conditions (BG-7)	Single use, epoch-bound and digest-bound; only from a human-approved runbook; verified by the supervisor or relay; spent write-ahead
Epochs and fencing	ADOPT (BG-8 to BG-11)	Authority belongs to one epoch. A trip, the fence and every boot open a new epoch. Fencing is enforced at the external target, because the orchestrator may be alive but partitioned. Rollback defence
Witness	ADOPT (BG-12 to BG-14)	Two independent signals (2-of-3 where a physical signal exists). The witness answers only "may emergency mode begin?". Independence is audited
Flight recorder	ADOPT (BG-15 to BG-18)	A separate chain written by the supervisor. Write-ahead ("no record, no action"), preallocated space, sealed and anchored
Recovery boundary	ADOPT as first-class states (OR-1 to OR-8)	The fence is one durable epoch write, done first. Reconciliation is mandatory. <code>SAFE_T0_ISOLATE</code> is a supervisor state. The human boundary is required at critical. HELD never times out. Probation has a closed checklist. Emergency authority has a hard maximum duration

Added in review:

1. **Two tiers.** A fixed outbound beacon (BG-5), which can come early. Acting on an external system (BG-A) is by definition class 3b, which is deferred to kernel v2.
2. **Every boot opens a new epoch.** Authority then dies even if power is cut mid-fence.
3. **A recovery panel that shows evidence, not verdicts.** "Trigger not observed for 14 min" instead of "cause cleared".

4.3 Costs stated deliberately (for review)

- **An unplanned restart during an emergency ends break glass.** Old envelopes are dead in the new epoch. The survival loop continues, and protection falls to the far side's own lost-link procedure. The alternative, allowing a crash-fenced emergency to trip again with its old envelopes, is weaker.
- **An emergency during probation gets no break glass.**
- **Break glass needs** the supervisor split (PD-72), the connector gate (PD-01.7), the activation register (PD-40) and an acknowledgement path for heartbeats. None is built.

4.4 Package B decision register

ID	Decision	Recommendation
PD-01.10	Premise, scope rule and two tiers	APPROVE; build BG-S after PD-01.3, PD-72 and PD-01.7; DEFER BG-A with 3b
PD-01.11	The three mechanisms as modified; a dormant relay rather than one started on demand	APPROVE
PD-01.12	Epochs and fencing; the conservative restart rule	APPROVE
PD-01.13	Witness and flight recorder	APPROVE
PD-01.14	The operator-gated recovery boundary as the only exit from any emergency <i>or survival</i> mode (this completes BF-6)	APPROVE

Questions for reviewers:

- **B1.** Is the external-only rule enforceable as specified (validator plus a connect-time check plus `single_path`)? Can you find an internal target it misses?
- **B2.** Is the conservative restart rule the right trade?
- **B3.** Can the witness be independent on a single DGX, or does it require separate hardware?
- **B4.** Does the state machine have a path by which authority survives a fence?

5. Package C: other open decisions that gate the next steps

These are compact. Each is set out in full in the file named.

ID	Decision	Recommendation	File
BF-1 to BF-6	Implementation conditions of the adopted Blind Forester ruling	APPROVE	PD-01-DAEMON-CLASSES.md §9b
PD-72	Supervisor/worker split as a precondition for any active class	APPROVE	INTEGRATION-REVIEW.md §7
PD-01.1, PD-01.2	The class ladder and invariants S-1 to S-8; every manifest declares a consequence level (critical refused)	APPROVE	PD-01-DAEMON-CLASSES.md §7
PD-01.3	Unlock 1b Sentinel: SENSE_DEGRADED at T/2, the spark-escalation/1 format, a read-only escalations command	APPROVE	same
PD-69	Limits are checked against each other (U-8), and every new timeout goes in a budget table	APPROVE	INTEGRATION-REVIEW.md §5
PD-71 (the part PD-76 does not cover)	A named worst-case fixture per observation type; a scaling series to derive N _{max}	APPROVE	same

ID	Decision	Recommendation	File
PD-76, open part	How the workspace and CI treat INCOMPLETE. Since r4.4, a direct reading is shown to be possible here without systemd, so INCOMPLETE should be rare	Require PASS where a direct reading is possible (Part 2 §2.4)	RECONCILIATION-V01-LINE.md §3
PD-75 to PD-81	Alias of the v0.1 line's PD-23 to PD-29, or renumber this repository's	Confirm the alias	same, §2
C-1 (with PD-81)	Battery exit codes. Here: PASS 0, INCOMPLETE 3, FAIL 1. v0.1 line: PASS 0, FAIL 3, INCOMPLETE 4	Adopt the v0.1 line's codes (0, 3, 4) in the batched contract change	same, §4
PD-64 to PD-68	Universal-contract rules: stack alignment; temporal honesty (S-8); stop and revocation; cumulative bounds; assurance of the assurance	APPROVE	ADJUDICATION-PLUG-AND-PLAY.md §9
A-1 to A-12	Amendments from the prior-art review (IEC 61784-3 black channel, NAMUR NE 107, ISA-18.2, IEC 62443, AUTOSAR, SOTIF, UL 4600, and others)	Rule on each	PRIOR-ART-REVIEW.md
Phase 1 go-ahead	An OnFailure= notifier; a diverse staleness checker that does not import the skeleton; a budget report. No contract change	APPROVE	INTEGRATION-REVIEW.md §4

6. Invariants every review should check against

Nothing in Packages A or B may weaken these:

- **S-1.** Never report health you do not have.
- **S-2.** Never record a guess as an observation. A capped or unsettled read is not an observation (HF-29, HF-33).
- **S-3.** Always surface blindness within T. Nothing pauses the blind clock.
- **S-4.** Stay confined. Higher classes gain request channels, never wider confinement.
- **S-5.** Authentication and handshakes never degrade.
- **S-6.** Proposals are not authority.
- **S-7.** Another system's safety protocol wins on its side.
- **S-8.** Never imply more temporal precision than you have.
- **U-7.** Evidence expires, and independence must be shown, not assumed.
- **U-8.** Limits are checked against each other, including over time (Package A's H03 finding).

7. Acceptance gates for this handoff

- Every item in Part 1 sections 3.5, 4.4 and 5, and every settled-ruling amendment in Part 2 section 2, is marked APPROVE, MODIFY, REJECT or DEFER.
- No Package B requirement is given an LTC number, and no LTC placeholder is consumed by Package B.
- No reviewer claim of a defect is accepted without a reproduction, a measurement or a code trace.
- No approval here activates a daemon, unlocks a class or changes the LTC version. Those remain the owner's Class C decisions.

Part 2: Decision brief

How to read the verdicts. A settled ruling can be kept, kept with an amendment (the decision stands; how it is built or worded changes), clarified (the decision stands; its scope is stated), or reversed. Every finding says what kind of evidence stands behind it:

- *measured*: run and recorded;
- *traced*: read in the code;
- *reasoned*: argued, not run.

1. Answer in brief

No settled ruling should be reversed. Each has held up, and one was confirmed by a new measurement. Each of the four does have one amendment or clarification worth making, and two of them fix real weaknesses in how the ruling was built:

Ruling	Verdict	What is worth revisiting	Evidence	Urgency
PD-63 Blind period	Keep, amend how it was built	SENSE_BLIND and a policy violation share exit code 78. When the output directory is unsafe, the ledger cannot be written and the exit code is the only signal left, so a security event looks like an operational one	Traced (spark_daemon/__init__.py: EXIT_POLICY = 78, EXIT_SENSE_BLIND = 78)	Medium; batch with the exit-code work (PD-35, C-1)
PD-70 Blindness survives restarts	Keep, amend how it was built	Blindness inherited across a restart is measured on the wall clock. If the clock steps back, inherited blindness reads as zero, and the slow restart loop that PD-70 fixed (HF-32) hides again. If it steps forward, a healthy daemon is judged blind	Traced (_seconds_since clamps a future timestamp to 0)	Medium-high; small and additive
PD-01.9 Blind Forester	Keep, clarify its scope	"Critical downstream payload" can be read as the critical consequence level, which PD-01.2 refuses until a certification path exists. The ruling should say that it does not unlock critical, and that the far side's own procedure (BF-5) is mandatory, not recommended	Reasoned	High, before any active class is designed
PD-76 Direct memory reading	Keep; amend the mechanism's wording	The ruling names cgroup v2's memory.peak from a systemd transient scope. This workspace has neither, yet a direct reading <i>can</i> be taken here, from a directly created cgroup (v1). The measurement confirms the ruling's premise: DB-14's basis read 128 MiB where the cgroup peaked at 208 MiB	Measured (evidence/r4.4/)	Medium; it removes most of the "INCOMPLETE in CI" problem

The r4.1 statement that "this workspace cannot take the reading" was wrong, and it is corrected where it appeared (RECONCILIATION-V01-LINE.md §3, the decision log).

2. The settled rulings, one by one

2.1 PD-63: the blind period (settled 2026-09-29)

Decided: a daemon that accepts no observation for T (blind_limit_seconds) stops with SENSE_BLIND, exit 78, and the unit never restarts it. T is declared in the manifest.

Learned since:

- **The exit code is shared.** EXIT_POLICY (an unsafe output directory, or a purity or policy violation: possibly an attack) and EXIT_SENSE_BLIND (the daemon cannot see: operational) are both 78. The ledger distinguishes them, but the unsafe-output-directory case is exactly the one where the ledger cannot be written, so the exit status is all that systemd, the journal and an OnFailure= notifier see. The two need different people at different urgency.
- **Why 78 was chosen:** the Observer's own table uses 78 for SENSE_BLIND (WBS 3.1), and this pattern aligns with the Observer voluntarily (PD-35).
- **A consequence of HF-33:** with a persistently over-capacity folder, dir-watch now fails every cycle and stops at T. That is the ruling working as intended: a capacity problem becomes a visible stop, not a silently wrong inventory. It is stated here so that it is chosen knowingly.

Options for the exit code:

Option	For	Against
A. Keep both at 78	No change; matches the Observer for SENSE_BLIND	A security stop and an operational stop are indistinguishable without the ledger, and the ledger may be the thing that failed
B. Move EXIT_POLICY to its own code; keep SENSE_BLIND at 78 (recommended)	Keeps the Observer alignment for the code the Observer defines. Separates security from operations where it matters most. RestartPreventExitStatus simply lists both	A contract major (exit codes are in the contract). Batch it with PD-35 and C-1
C. Move SENSE_BLIND instead	—	Breaks the Observer alignment

Options for where T is set (a lower priority):

Option	For	Against
A. In the manifest (as ruled; recommended for now)	Hashed into DAEMON_START and bound to activation; changing it is visible	Deploying the same daemon on two hosts with different T needs two manifests
B. The manifest declares a range; the activation register (PD-40) picks T within it and records the choice	Per-deployment tuning without re-approving the daemon	Needs the register, which is not built. Two places to look for one number

Recommendation: keep PD-63. Adopt exit-code option B in the batched contract change. Revisit where T is set when PD-40 is built.

2.2 PD-70: blindness survives restarts (settled 2026-09-30)

Decided:

- a heartbeat every $T/2$;
- at start-up, the blind clock begins at the ledger's latest evidence of an accepted cycle;
- one reacquisition cycle per restart, so that no daemon is locked out.

Learned since: the time since that evidence is measured on the **wall clock** (`_seconds_since`, which returns 0 when the timestamp lies in the future). Two failure cases follow from the code:

- **The clock steps back** (an NTP correction after a fast real-time clock, or a manual change). The last accepted timestamp then lies "in the future", inherited blindness reads as 0, and every restart starts a fresh countdown. This brings HF-32 back: a slow restart loop hides blindness indefinitely, which is the silent outage PD-70 was adopted to prevent.
- **The clock steps forward** (a real-time clock reset at boot, then NTP). A healthy daemon inherits a large blindness and gets only its single reacquisition cycle. That is safe, but it is a false stop.

Within one boot there is a clock that no wall-clock change moves: Linux's CLOCK_BOOTTIME, which is shared by all processes since boot and counts suspended time. Across a reboot no local clock is reliable. The Observer's own rule (D-7) already keeps monotonic time within a process.

Option	For	Against
A. Keep the wall clock (as built)	Simple	A backward step silently brings back the defect PD-70 fixed
B. Record boot_id and CLOCK_BOOTTIME with every piece of acceptance evidence (heartbeat, start, stop). Within the same boot, measure on CLOCK_BOOTTIME. Across boots, use the wall clock; if it went backward, or cannot be trusted, assume the worst (blindness at the limit, so one reacquisition cycle). Record clock_basis (recommended)	Immune to clock steps within a boot, which is where restart loops happen. Honest across boots (S-8), and errs towards a check, never towards silence	Additive fields in reserved records (a contract minor). Blindness measured from the last heartbeat instead of the last event can over-state it by up to T/2, in the safe direction
C. Always assume the worst across restarts	Simplest safe rule	Every restart costs its one reacquisition cycle; a daemon on a flaky input stops more often

Recommendation: keep PD-70 and build option B. A test can inject a backward step through the ledger's timestamps; it should fail on the current code and pass after the change.

2.3 PD-01.9: the "Blind Forester" protocol (settled 2026-09-30, adopted as core pattern)

Decided: an active daemon whose stopping would drop a critical downstream payload does not fail closed when blind. It shifts into a pre-validated, degraded survival loop and signals distress. A restart resumes that loop. For observers the survival action stays "fail closed". The ruling superseded, in part, the earlier boundary that no DGX process may sit in a loop that keeps a person safe.

Learned since:

- 1. **The word "critical" is ambiguous.** It can be read as the consequence level *critical*, which PD-01.2 refuses "until a certification path exists". Read that way, the ruling would unlock life-safety duty for a general-purpose AI host with no certification. That is almost certainly not intended, but the text does not rule it out.
- 2. **BF-5 (the far side keeps its own procedure) is still only a recommendation.** It is the condition that keeps a person safe if the whole DGX fails: its kernel, power or disk.
- 3. **Break Glass (Package B) separates survival mode from emergency authority.** The mode continues across a restart; the authority does not. The Blind Forester ruling should say which one it covers: the mode.
- 4. **Exit from survival mode.** BF-6 requires revalidation. PD-01.14 proposes that the operator-gated recovery path be the only exit.
- 5. **Unit rules.** Today the unit never restarts exit 78. An active daemon that exits 78 when blind would drop its payload, the opposite of the ruling. BF-1 (the loop runs in the supervisor, which does not exit) resolves this, so the supervisor split (PD-72) is a hard prerequisite, not an option.

Option	For	Against
A. Keep as ruled; conditions pending	No change	The scope ambiguity stays open until someone designs against it
B. Keep, and add a scope clarification (recommended): the protocol applies to active classes at standard and elevated consequence; it does not unlock <i>critical</i> , which stays refused under PD-01.2; BF-5 is mandatory wherever a person could be	Keeps the owner's intent. Closes the reading that would put an uncertified host in a life-safety loop. Aligns with Package B	The owner may have intended <i>critical</i> to be reachable. If so, that needs its own decision with a certification path

Option	For	Against
harmed; it governs survival <i>mode</i> , never emergency <i>authority</i> ; the supervisor split is a prerequisite		
C. Narrow the ruling to elevated only, or return to the old boundary	The simplest safety case	Loses a first line of defence the owner deliberately adopted

Recommendation: keep PD-01.9 with the clarification in option B, and confirm BF-1 to BF-6, with BF-5 as mandatory.

2.4 PD-76 (= the v0.1 line's PD-24(a)): a direct memory reading (settled 2026-09-29 in that line)

Decided: the memory check needs a direct reading of `cgroup memory.peak` from a fresh transient scope containing the daemon and every child. A process-RSS reading is a proxy that never decides, and an RSS-only run is INCOMPLETE.

Learned since (measured, evidence/r4.4/). Running as root on this workspace, with no systemd user manager, a child memory cgroup was created under the shell's own cgroup. A process was moved into it that held 80 MiB while its child allocated 120 MiB:

Basis	Reading
DB-14 today: <code>max(daemon, largest child)</code>	128 MiB
<code>cgroup v1 memory.max_usage_in_bytes</code> , daemon plus every child	208 MiB

- **The premise is confirmed.** DB-14's basis reads about 38 % low whenever a daemon holds memory while a child runs, and `MemoryMax=` enforces the cgroup figure.
- **The mechanism in the ruling is narrower than necessary.** This workspace uses cgroup v1 (no `memory.peak`) and has no systemd user manager (no transient scope), yet a direct reading was taken. So the r4.1 conclusion, that every battery run here would be INCOMPLETE, was wrong.

Option	For	Against
A. Keep the wording (v2 <code>memory.peak</code> from a systemd transient scope)	Exactly what the target, the DGX on cgroup v2 with systemd, will use	INCOMPLETE on every host without both, including this one, although a direct reading is possible
B. Keep the ruling; widen the mechanism (recommended): "a direct cgroup peak for a fresh cgroup holding the daemon and every child: v2 <code>memory.peak</code> , or v1 <code>memory.max_usage_in_bytes</code> , created by a systemd transient scope or directly where the battery may create cgroups. The report records the cgroup version and how the cgroup was created. INCOMPLETE only where no direct reading is possible"	Same substance. INCOMPLETE becomes rare instead of universal, and CI can require PASS where it runs as root or with delegated cgroups	v1 and v2 account for memory differently (page cache, kernel memory), so readings from different versions are not interchangeable. Compare each only against a limit on the same kind of host. The battery needs cgroup write access
C. Go back to RSS	—	Measured 38 % low; reversing would contradict the evidence

Recommendation: keep PD-76 and adopt option B's wording. Build DB-19 on it, and record the cgroup version in the report.

3. The open decisions: choices, pros and cons

Each row gives the realistic options. The recommended option is marked **(rec.)**.

3.1 Package A: the Local Transport Contract

Decision	Options	Pros	Cons
PD-82 Where the LTC sits	Bind at declared connections (rec.) / embed in the pattern / keep separate	Bind: one transport meaning, observers untouched, the gate's tests <i>are</i> the LTC suite	Bind: the seam is designed before its first use. Embed: couples version lines and pulls daemon constants into the LTC. Separate: each class re-derives transport rules
PD-83 Truncated reads	The framework raises TooLarge by default; partial results only by opt-in (rec.) / a battery check only / leave it to authors	Framework: closes the API shape behind HF-29 and HF-33 for every author	Framework: a contract major. Battery-only: catches only what a fixture exercises. Authors: two defects already show this fails
PD-84 Register vocabulary	Adopt the LTC binding fields, with H02's modifications (rec.) / keep this pattern's own names	One vocabulary for daemons and providers	Ties the register's design to an unadjudicated LTC; mitigated because the fields are semantic, not a wire format
PD-85 Pacing	Declared field; deterministic spread by default (rec.) / keep random jitter (PD-21) / none	Deterministic: reproducible and testable; fixed-phase workloads get no hidden jitter	Deterministic: a fixed phase per daemon, so two daemons that hash close together stay close (mitigated by also hashing the host identity). Random: not reproducible
PD-86 Retry bounds	Bounded over time, checked across layers (rec.) / per-layer bounds only	Closes the HF-28 class of loop	Every retrying layer must declare a lifetime bound or a terminal state

3.2 Package B: Break Glass and recovery

Decision	Options	Pros	Cons
PD-01.10 Tiers	Beacon first, acting with class 3b (rec.) / build both together / defer all	Beacon first: useful early, cannot act, proves the recovery states before anything depends on them	Beacon first: two build stages. Both together: waits for kernel v2. Defer: no distress signal when the control plane is lost
PD-01.11 Egress relay	Always present and dormant (rec.) / started only on a trip	Dormant: nothing privileged has to start a unit at the worst moment	Dormant: a standing egress permission, although policy-gated and narrowly scoped. On-trip: no standing egress, but needs a privileged starter
PD-01.11 Survival socket	The owner's revised socket (rec.) / no socket (physical console only) / the original unauthenticated socket	Revised: zero friction for declared peers; exists only during an emergency	No socket: the smallest attack surface, but no local controller path. Unauthenticated: rejected, because it can be squatted and contradicts S-5
PD-01.12 Restart during an emergency	Break glass ends and survival continues (rec.) / allow re-arming with old envelopes	Conservative: authority never outlives its epoch	Conservative: after a crash, only the far side's procedure protects the payload. Permissive: stolen or stale envelopes live longer
PD-01.13 Witness	Separate hardware where a person could be harmed; a separate process tree otherwise (rec.) / always same-host / always separate hardware	Matches independence to consequence (U-7, BM-3)	Separate hardware costs money and needs integration. Same-host shares failure modes
PD-01.14 Recovery gate	Operator required at critical, declared per deployment below that (rec.) / always operator / always automatic	A human boundary where it matters; automatic where a stop costs only time	Always-operator: slow recovery for trivial daemons. Always-automatic: broad authority returns because a heartbeat did

3.3 Package C: other decisions that gate the next steps

Decision	Options	Pros	Cons
BF-1 to BF-6	Confirm all, BF-5 mandatory (rec.) / confirm a subset	Makes the adopted ruling buildable and safe	None material; each condition follows from a recorded invariant
PD-72 Supervisor/worker split	A precondition for any active class; build with the first one (rec.) / build now / never	Required by BF-1 and LTC-14; building it later avoids premature complexity	Build now: effort with no active class to use it. Never: the Blind Forester cannot be built
PD-01.1 to PD-01.3 Classes, consequence field, Sentinel	Approve; ship 1b next (rec.) / keep 1a only	1b closes silent outages at the source; a consequence field makes risk explicit	A contract change; alert fatigue unless ISA-18.2's rules (A-5) come with it
PD-69 Limits checked against each other	Approve (rec.) / ad hoc	Four defects came from limits checked alone (HF-28, HF-30, HF-31, HF-32)	A budget table to maintain
PD-71 (remainder) Worst-case fixtures, N_max	Approve with DB-19 (rec.) / defer	Evidence where the risk is, not on one small fixture	More battery time
PD-75 to PD-81 Numbering	Alias (rec.) / renumber this repository	Alias: no document rewritten	Renumber: churn across many documents; either works if chosen once
C-1 Battery exit codes	Adopt 0/3/4 (PASS/FAIL/INCOMPLETE) in the batched change (rec.) / keep 0/1/3 / map	One convention with SPS-1	A breaking change for anyone scripting on today's codes; batch it once
INCOMPLETE in CI	Require PASS where a direct reading is possible; allow INCOMPLETE with a stated environment reason elsewhere (rec.) / always require PASS / always allow	Honest, and after §2.4 rarely needed	Always PASS: fails on hosts that cannot measure. Always allow: hides real gaps
Phase 1 (OnFailure=notifier, diverse staleness checker, budget report)	Go (rec.) / wait	No contract change; each item closes a monitoring gap	Small effort

4. Suggested order of rulings

1. **Clarify PD-01.9** (§2.3). It frames everything designed for active classes.
2. **Confirm BF-1 to BF-6 and the Phase 1 go-ahead.** Neither changes the contract.
3. **Amend PD-70's clock basis** (§2.2). Small, additive, and it closes a regression path for an owner ruling.
4. **Adopt PD-76's wider mechanism** (§2.4), and with it the CI rule for INCOMPLETE. Then build DB-19.
5. **The batched contract change (4.0.0):**
 - PD-63's exit-code split;
 - C-1 and PD-81 (battery exit codes);
 - PD-35, PD-39 and PD-49;
 - PD-83 (truncated reads);
 - PD-85 (pacing);
 - PD-01.2 (consequence field).
6. **PD-82, PD-84 and PD-86**, as design rules for the register and the gate.
7. **Package B** (PD-01.10 to PD-01.14), after PD-72 and PD-01.7 are ruled on.

Part 3: Markup and document map

Markup table

Mark each row. Settled rulings (Part 2 §2): KEEP, AMEND, CLARIFY or REVERSE. Open decisions: APPROVE, MODIFY, REJECT or DEFER. Evidence type: reproduced, measured, traced or opinion.

Item	Where	Verdict	Evidence type	Notes
PD-63: exit-code split	Part 2 §2.1			
PD-63: where T is set	Part 2 §2.1			
PD-70: clock basis	Part 2 §2.2			
PD-01.9: scope clarification	Part 2 §2.3			
PD-76: mechanism wording, and INCOMPLETE in CI	Part 2 §2.4			
PD-82: bind, do not embed	Part 1 §3.5, Part 2 §3.1			
PD-83: truncated reads raise	Part 1 §3.5, Part 2 §3.1			
PD-84: register vocabulary	Part 1 §3.5, Part 2 §3.1			
PD-85: declared pacing	Part 1 §3.5, Part 2 §3.1			
PD-86: retry bounds over time	Part 1 §3.5, Part 2 §3.1			
LTC-H01 to H03: this review's additions	Part 1 §3.4			
PD-01.10: tiers and scope	Part 1 §4.4, Part 2 §3.2			
PD-01.11: relay and socket	Part 1 §4.4, Part 2 §3.2			
PD-01.12: epochs and restart rule	Part 1 §4.4, Part 2 §3.2			
PD-01.13: witness and flight recorder	Part 1 §4.4, Part 2 §3.2			
PD-01.14: recovery boundary	Part 1 §4.4, Part 2 §3.2			
BF-1 to BF-6	Part 1 §5, Part 2 §3.3			
PD-72: supervisor/worker split	Part 1 §5, Part 2 §3.3			
PD-01.1 to PD-01.3	Part 1 §5, Part 2 §3.3			
PD-69, PD-71	Part 1 §5, Part 2 §3.3			
PD-75 to PD-81, C-1	Part 1 §5, Part 2 §3.3			
PD-64 to PD-68, A-1 to A-12	Part 1 §5			
Phase 1 go-ahead	Part 1 §5, Part 2 §3.3			
Seams N-1 to N-14: any not covered by the design	Reviewer focus			

Appendix: document map

File	Contents
README.md	Overview, the decision list, the decision log, the revision history
HANDOFF-REVIEW-PACKAGE.md	Part 1 of this document, on its own
DECISION-BRIEF.md	Part 2 of this document, on its own
HARDENING.md	HF-01 to HF-33: each defect, its reproduction, its fix and its test
DAEMON-CONTRACT.md	The author contract, the candidate envelope, evidence lanes
PD-01-DAEMON-CLASSES.md	Classes, consequence levels, invariants, blind modes, Blind Forester, Break Glass summary

File	Contents
BREAK-GLASS.md	Package B in full
LTC-INTEGRATION-REVIEW.md	Package A in full
INTEGRATION-REVIEW.md	Limits checked against each other, the integration plan, owner rulings, the hardening review
ADJUDICATION-PLUG-AND-PLAY.md	The L0 to L4 contract stack, kernel v2 agenda, universal-contract rules U-1 to U-7
PRIOR-ART-REVIEW.md	Industrial, flight, automotive, robotics and AI-agent prior art; A-1 to A-12
RECONCILIATION-V01-LINE.md	The parallel v0.1 line: alias table, conflicts, the owner's ruling PD-76
evidence/	Reproductions and measurements by revision